

Stochastic Reconfiguration in Quantum Monte Carlo

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- imaginary-time evolution and variational projection,
- tangent-space expansion of the trial state,
- derivation of the linear system $S \delta \alpha = -\delta \tau \mathbf{g}$,
- interpretation of the covariance matrix S ,
- relation to the natural gradient.

Setup and notation

We consider a variational wave function

$$|\Psi(\alpha)\rangle, \quad \alpha = (\alpha_1, \alpha_2, \dots, \alpha_p),$$

depending on p real parameters.

The normalized state is

$$|\bar{\Psi}(\alpha)\rangle = \frac{|\Psi(\alpha)\rangle}{\sqrt{\langle\Psi(\alpha)|\Psi(\alpha)\rangle}}.$$

We wish to optimize the parameters such that the energy

$$E(\alpha) = \frac{\langle\Psi(\alpha)|H|\Psi(\alpha)\rangle}{\langle\Psi(\alpha)|\Psi(\alpha)\rangle}$$

is minimized.

Imaginary-time evolution as a route to the ground state

The exact imaginary-time evolution of a state is

$$|\Psi(\tau)\rangle = e^{-\tau H} |\Psi(0)\rangle .$$

Expanding the initial state in energy eigenstates,

$$|\Psi(0)\rangle = \sum_n c_n |n\rangle , \quad H |n\rangle = E_n |n\rangle ,$$

gives

$$|\Psi(\tau)\rangle = \sum_n c_n e^{-\tau E_n} |n\rangle .$$

If $c_0 \neq 0$, then for large τ ,

$$|\Psi(\tau)\rangle \propto e^{-\tau E_0} |0\rangle ,$$

so imaginary-time propagation filters out the ground state.

Why a variational projection is needed

In Variational Monte Carlo we do not evolve an arbitrary state in the full Hilbert space. Instead, we restrict ourselves to the variational manifold

$$\mathcal{M} = \{|\Psi(\alpha)\rangle \mid \alpha \in \mathbb{R}^p\}.$$

The exact state

$$(1 - \delta\tau H) |\Psi(\alpha)\rangle$$

will in general *not* belong to $\mathcal{M}$.

Idea of stochastic reconfiguration:

- apply one infinitesimal imaginary-time step,
- project the resulting state back onto the variational manifold,
- interpret the projection as a parameter update $\alpha \rightarrow \alpha + \delta\alpha$.

First-order imaginary-time step

For a small time step $\delta\tau$,

$$e^{-\delta\tau H} = 1 - \delta\tau H + \mathcal{O}(\delta\tau^2).$$

Thus the exact propagated state is approximated by

$$|\Psi'_{\text{exact}}\rangle \approx (1 - \delta\tau H) |\Psi(\alpha)\rangle.$$

A convenient first-order, norm-preserving form is

$$|\Psi'_{\text{exact}}\rangle \approx [1 - \delta\tau(H - E(\alpha))] |\Psi(\alpha)\rangle.$$

Subtracting $E(\alpha)$ removes the trivial component parallel to $|\Psi\rangle$.

Why subtract $E(\alpha)$?

Let

$$E = \frac{\langle \Psi | H | \Psi \rangle}{\langle \Psi | \Psi \rangle}.$$

Then

$$\langle \Psi | (H - E) | \Psi \rangle = 0.$$

Hence the vector

$$(H - E) | \Psi \rangle$$

is orthogonal to $|\Psi\rangle$ in the sense relevant for norm changes to first order.

Therefore,

$$[1 - \delta\tau(H - E)] | \Psi \rangle$$

moves the state in a direction tangent to projective Hilbert space, removing the trivial norm-rescaling component.

Variational state after a parameter update

We now expand the variational state at shifted parameters:

$$|\Psi(\alpha + \delta\alpha)\rangle = |\Psi(\alpha)\rangle + \sum_k \delta\alpha_k \frac{\partial}{\partial\alpha_k} |\Psi(\alpha)\rangle + \mathcal{O}(\delta\alpha^2).$$

Define the tangent vectors

$$|\Psi_k\rangle \equiv \frac{\partial}{\partial\alpha_k} |\Psi(\alpha)\rangle.$$

Then

$$|\Psi(\alpha + \delta\alpha)\rangle = |\Psi\rangle + \sum_k \delta\alpha_k |\Psi_k\rangle + \mathcal{O}(\delta\alpha^2).$$

Projection principle: minimize the distance

We compare

$$|\Psi'_{\text{var}}\rangle = |\Psi\rangle + \sum_k \delta\alpha_k |\Psi_k\rangle$$

with

$$|\Psi'_{\text{exact}}\rangle = [1 - \delta\tau(H - E)] |\Psi\rangle.$$

Define the mismatch vector

$$|\Delta\rangle = |\Psi'_{\text{var}}\rangle - |\Psi'_{\text{exact}}\rangle.$$

To first order,

$$|\Delta\rangle = \sum_k \delta\alpha_k |\Psi_k\rangle + \delta\tau(H - E) |\Psi\rangle.$$

The SR update is obtained by minimizing

$$\mathcal{D} = \langle\Delta|\Delta\rangle.$$

Expansion of the distance functional

Using

$$|\Delta\rangle = \sum_k \delta\alpha_k |\Psi_k\rangle + \delta\tau(H - E)|\Psi\rangle,$$

we obtain

$$\mathcal{D} = \left(\sum_i \delta\alpha_i \langle\Psi_i| + \delta\tau \langle\Psi| (H - E) \right) \left(\sum_j \delta\alpha_j |\Psi_j\rangle + \delta\tau(H - E)|\Psi\rangle \right).$$

Expanding and keeping terms up to first order in $\delta\tau$,

$$\mathcal{D} = \sum_{ij} \delta\alpha_i \delta\alpha_j \langle\Psi_i|\Psi_j\rangle + 2\delta\tau \sum_i \delta\alpha_i \operatorname{Re} \langle\Psi_i| (H - E) |\Psi\rangle + \mathcal{O}(\delta\tau^2).$$

Stationarity condition

Differentiate $\mathcal{D}$ with respect to $\delta\alpha_i$:

$$\frac{\partial \mathcal{D}}{\partial(\delta\alpha_i)} = 2 \sum_j \langle \Psi_i | \Psi_j \rangle \delta\alpha_j + 2\delta\tau \operatorname{Re} \langle \Psi_i | (H - E) | \Psi \rangle .$$

Setting this equal to zero gives

$$\sum_j \langle \Psi_i | \Psi_j \rangle \delta\alpha_j = -\delta\tau \operatorname{Re} \langle \Psi_i | (H - E) | \Psi \rangle .$$

This is the projected imaginary-time equation in the tangent basis $\{|\Psi_i\rangle\}$.

Normalized-state formulation

To remove the redundant component parallel to $|\Psi\rangle$, introduce

$$|\overline{\Psi}\rangle = \frac{|\Psi\rangle}{\sqrt{\langle\Psi|\Psi\rangle}}.$$

Define normalized tangent vectors:

$$|\overline{\Psi}_k\rangle = \frac{\partial}{\partial\alpha_k} |\overline{\Psi}\rangle.$$

Then the projected equation becomes

$$\sum_j \langle\overline{\Psi}_i|\overline{\Psi}_j\rangle \delta\alpha_j = -\delta\tau \langle\overline{\Psi}_i|(H-E)|\overline{\Psi}\rangle.$$

This form leads directly to the SR metric matrix.

Logarithmic derivatives in configuration space

Let x denote a many-body configuration and write the wave function as $\Psi(x)$.

Define the logarithmic derivative operators

$$O_k(x) = \frac{1}{\Psi(x)} \frac{\partial \Psi(x)}{\partial \alpha_k} = \frac{\partial \ln \Psi(x)}{\partial \alpha_k}.$$

Then

$$\frac{\partial \Psi(x)}{\partial \alpha_k} = O_k(x) \Psi(x).$$

For Monte Carlo sampling with probability

$$P(x) = \frac{|\Psi(x)|^2}{\langle \Psi | \Psi \rangle},$$

expectation values are

$$\langle A \rangle = \sum_x P(x) A(x)$$

(or integrals in the continuous case).

Deriving the metric matrix S_{ij}

Using normalized states, one finds

$$\langle \bar{\Psi}_i | \bar{\Psi}_j \rangle = \langle O_i^* O_j \rangle - \langle O_i^* \rangle \langle O_j \rangle.$$

For real wave functions and real parameters this becomes

$$S_{ij} = \langle O_i O_j \rangle - \langle O_i \rangle \langle O_j \rangle.$$

Thus S_{ij} is the covariance matrix of the logarithmic derivatives.

Interpretation:

- S measures how parameter changes alter the normalized state,
- S is the metric tensor on the variational manifold,
- S is positive semidefinite.

Deriving the generalized force g_i

The right-hand side is

$$\langle \bar{\Psi}_i | (H - E) | \bar{\Psi} \rangle.$$

Define the local energy

$$E_{\text{loc}}(x) = \frac{(H\Psi)(x)}{\Psi(x)}.$$

Then

$$g_i = \text{Re} [\langle \bar{\Psi}_i | (H - E) | \bar{\Psi} \rangle] = \langle O_i E_{\text{loc}} \rangle - \langle O_i \rangle \langle E_{\text{loc}} \rangle$$

for real-valued states.

Thus g_i is the covariance between the logarithmic derivative and the local energy.

Final SR linear system

Putting everything together yields

$$\sum_j S_{ij} \delta\alpha_j = -\delta\tau g_i,$$

with

$$S_{ij} = \langle O_i O_j \rangle - \langle O_i \rangle \langle O_j \rangle,$$

and

$$g_i = \langle O_i E_{\text{loc}} \rangle - \langle O_i \rangle \langle E_{\text{loc}} \rangle.$$

In vector form:

$$S \delta\alpha = -\delta\tau \mathbf{g}, \quad \delta\alpha = -\delta\tau S^{-1} \mathbf{g}.$$

Why this differs from ordinary gradient descent

Ordinary gradient descent updates

$$\delta \alpha_{\text{GD}} = -\eta \mathbf{g}.$$

SR instead uses

$$\delta \alpha_{\text{SR}} = -\eta S^{-1} \mathbf{g}.$$

The inverse metric S^{-1} rescales and mixes parameter directions according to the geometry of the wave-function manifold.

Consequences:

- reduced sensitivity to redundant parameterizations,
- better conditioning,
- close relation to imaginary-time evolution.

Connection to the natural gradient

The squared distance between nearby normalized states is

$$ds^2 = \sum_{ij} S_{ij} d\alpha_i d\alpha_j.$$

Thus S defines a Riemannian metric in parameter space.

Steepest descent in this metric gives

$$\delta\alpha \propto -S^{-1}\mathbf{g},$$

which is exactly the **natural gradient** update.

Therefore SR is the natural-gradient method specialized to variational quantum states.

Regularization and practical solution

In practice S may be singular or ill-conditioned.

A standard fix is diagonal regularization:

$$S \rightarrow S + \lambda I, \quad \lambda > 0.$$

Then solve

$$(S + \lambda I) \delta \alpha = -\delta \tau \mathbf{g}.$$

Interpretation:

- stabilizes inversion,
- damps noisy directions,
- interpolates between SR and ordinary gradient descent for large λ .

Algorithmic summary

At each iteration:

① Sample configurations x from $P(x) = |\Psi(x)|^2 / \langle \Psi | \Psi \rangle$.

② Compute logarithmic derivatives $O_k(x)$.

③ Compute local energies $E_{\text{loc}}(x)$.

④ Estimate

$$S_{ij} = \langle O_i O_j \rangle - \langle O_i \rangle \langle O_j \rangle,$$

$$g_i = \langle O_i E_{\text{loc}} \rangle - \langle O_i \rangle \langle E_{\text{loc}} \rangle.$$

⑤ Solve

$$(S + \lambda I) \delta \alpha = -\delta \tau \mathbf{g}.$$

⑥ Update $\alpha \leftarrow \alpha + \delta \alpha$.

Summary of the derivation

- Imaginary-time evolution suggests the optimal direction toward the ground state.
- The variational manifold restricts admissible states to $|\Psi(\alpha)\rangle$.
- Projecting the infinitesimal imaginary-time step onto the tangent space yields

$$S \delta \alpha = -\delta \tau \mathbf{g}.$$

- S is the covariance matrix of logarithmic derivatives.
- $\mathbf{g}$ is the covariance of logarithmic derivatives with the local energy.
- SR is both:
 - a projected imaginary-time evolution,
 - and a natural-gradient optimization method.